

MicrobeTrace Network Display Comparison

Cytoscape WebGL vs Sigma - plain-language summary of current benchmark evidence

Sigma is the leading option for feature-rich networks: it is smoother and uses less complete browser and graphics memory. Cytoscape remains faster for immediate selection and group collapse.

60.1 FPS

Sigma pan speed

1.20 sec

Sigma network ready

554 MiB

Sigma added Chrome
memory

2.2 ms

Cytoscape selection

Primary test environment: Chrome 152 | Five runs per renderer | Complete memory measured September 14, 2026

Plain-language summary

This report compares two tuned options for displaying MicrobeTrace networks: Cytoscape WebGL and Sigma. Both were tested with the same data, node positions, visible connections, and feature requirements.

Sigma is the leading option for the feature-rich network display. It was smoother while moving through the network, opened the feature-rich view sooner, and used less complete browser and graphics memory. Cytoscape was faster for immediate selection and group collapse.

The scientific grouping rules should remain part of MicrobeTrace instead of belonging to either display library. That lets the same case, specimen, and sequence relationships support networks, tables, filters, maps, statistics, exports, and saved sessions.

What we tested

Test network	Size	What it represents	Why it matters
Distance network	1,600 nodes; 537 connections	A committed MicrobeTrace genetic-distance dataset	Normal, relatively sparse network
Distance plus epidemiologic links	1,600 nodes; 596 connections	The same network with 59 added epidemiologic links	Mixed connection types
Feature-rich network	1,600 nodes; 3,200 connections	40 groups, donut nodes, QC flags, uncertainty, and locations	Future MicrobeTrace display needs

The main tests used Chrome 152. Each result is the middle value from five runs, followed by the lowest and highest values in brackets. A test failed if the requested WebGL display stopped working or if required nodes or connections were missing.

Feature-rich network results

Measure	Cytoscape WebGL	Sigma	Better result	What a user would notice
Network ready	3.65 sec [3.55-4.02]	1.20 sec [1.16-2.68]	Sigma	The view becomes usable sooner
Pan smoothness	40.2 FPS [35.7-43.2]	60.1 FPS [60.1-60.2]	Sigma	Smoother movement
Zoom smoothness	45.9 FPS [42.9-60.1]	60.3 FPS [60.2-60.3]	Sigma	Smoother zooming
Selection response	2.2 ms [1.9-2.4]	13.2 ms [12.3-15.6]	Cytoscape	Both feel immediate; Cytoscape updates faster
Export	11.9 ms [11.3-41.4]	8.8 ms [8.2-10.7]	Sigma	Both are fast; Sigma is more consistent
Collapse 40 groups	60.0 ms [52.4-71.9]	93.4 ms [88.5-102.1]	Cytoscape	Both feel immediate
Reopen all groups	316.6 ms [252.9-487.6]	191.2 ms [181.4-211.5]	Sigma	Grouped data returns sooner

Sigma stayed at roughly 60 FPS for every pan and zoom run. This was the clearest performance difference between the two tuned implementations.

Complete browser and graphics memory

These measurements use five fresh Chrome processes per renderer on Windows. Each run tracks the complete seven-process Chrome group: the browser, page renderers, utilities, GPU process, and crash handler. Readings were taken before loading the dataset and at four later checkpoints. The largest checkpoint is called "peak observed"; it is not a continuous maximum.

Complete Chrome memory	Cytoscape WebGL	Sigma	Difference
Starting working memory	1,029.8 MiB [1,026.8-1,047.3]	1,041.2 MiB [1,026.5-1,073.9]	Similar starting point
Added working memory	1,153.6 MiB [1,034.6-1,184.6]	553.8 MiB [295.1-626.5]	Sigma used about 600 MiB less
Final working memory	2,186.1 MiB [2,061.4-2,213.5]	1,595.0 MiB [1,369.0-1,667.5]	Sigma used about 27% less overall
Added private memory	1,131.2 MiB [1,010.7-1,160.9]	531.4 MiB [266.0-601.8]	Sigma used about 53% less added memory
Peak observed working memory	2,186.1 MiB [2,076.9-2,213.5]	1,596.8 MiB [1,483.3-1,667.5]	Sigma used about 589 MiB less

Graphics memory	Cytoscape WebGL	Sigma	Difference
Starting graphics memory	125.4 MiB [121.9-138.5]	141.1 MiB [139.7-162.6]	Similar starting range
Added graphics memory	510.7 MiB [390.8-529.8]	328.2 MiB [117.6-333.3]	Sigma used about 183 MiB less
Final graphics memory	649.2 MiB [512.6-657.9]	470.9 MiB [280.2-474.3]	Sigma used about 28% less overall
Peak observed graphics memory	656.1 MiB [535.5-684.3]	470.9 MiB [282.2-474.3]	Sigma used about 185 MiB less

This computer uses shared graphics memory, so its dedicated GPU reading was zero. The table reports the shared memory actively assigned to Chrome. We also measured JavaScript memory separately: Cytoscape added 83.1 MiB [82.6-84.5], while Sigma added 41.2 MiB [41.1-42.5]. JavaScript memory is useful for diagnosis, but it is only one part of the complete browser total.

Representative MicrobeTrace networks

The representative networks have fewer visible connections than the feature-rich network. The two display systems performed similarly for opening and moving around these networks.

Network	Measure	Cytoscape WebGL	Sigma	Plain-language result
Distance	Network ready	1.22 sec [0.92-1.43]	1.19 sec [0.81-2.16]	No meaningful typical difference
Distance	Pan smoothness	55.3 FPS [3.7-60.2]	53.0 FPS [47.2-60.2]	Similar medians; Cytoscape had one large pause
Distance	Zoom smoothness	55.6 FPS [46.1-60.4]	54.1 FPS [46.8-60.2]	Similar
Distance	Selection response	0.8 ms [0.7-1.0]	6.4 ms [5.7-6.9]	Cytoscape is faster; both feel immediate
Distance plus epidemiologic links	Network ready	1.23 sec [0.80-1.82]	1.31 sec [0.93-3.33]	No meaningful typical difference
Distance plus epidemiologic links	Pan smoothness	51.9 FPS [43.5-60.1]	49.5 FPS [21.0-58.3]	Similar medians; both had occasional pauses
Distance plus epidemiologic links	Zoom smoothness	51.9 FPS [44.6-60.5]	49.2 FPS [33.3-59.0]	Similar
Distance plus epidemiologic links	Selection response	0.8 ms [0.7-1.0]	6.3 ms [6.2-16.5]	Cytoscape is faster; both feel immediate

Both systems showed all required connections. The important performance difference appears when richer visual features and more connections are added.

Feature and reliability scorecard

User need	Cytoscape WebGL	Sigma	Current result
Mixed-value donut nodes	Added Canvas drawing layer	Custom WebGL symbols	Both pass; Sigma is smoother at feature scale
Groups and cluster outlines	Built-in parent/child groups plus shared collapse rules	Custom outlines plus shared collapse rules	Membership, dragging, selection, and statistics pass
QC and uncertainty markers	Added Canvas drawing layer	Custom WebGL symbols	Both pass for all 1,600 nodes
Collapse and reopen groups	Shared MicrobeTrace rules	Same rules	Data, selections, and statistics are preserved
Geographic positions	Shared positions and reference overlay	Same	Positions, group centers, export, and restore pass
Selection and highlighting	Fast built-in selection	Targeted partial update	Mouse, keyboard, group, neighbor, and cross-view behavior pass
Export and saved sessions	Complete image/SVG export and session restore	Same	Expanded and collapsed views restore correctly
Accessibility	Shared keyboard controls and text descriptions	Same	Automated checks pass; manual screen-reader check remains
WebGL interruption	Switches to a Canvas display	Recreates the WebGL display	Both retain the network and selection state
Browser coverage	Chrome full run; Edge secondary check	Same	Both current target browsers pass

Both systems create complete PNG and SVG exports. The SVG contains the complete image, but its individual shapes are not fully editable. That only matters if editable artwork is a requirement.

Grouping should belong to MicrobeTrace

Cytoscape can store parent/child relationships inside its display library. Sigma cannot do that by itself. MicrobeTrace does not need to choose a display library based only on this difference.

MicrobeTrace now owns the grouping rules and asks each library to draw the result. The same case, specimen, and sequence relationships can therefore support tables, filters, maps, statistics, exports, and saved sessions.

MicrobeTrace controls	Cytoscape displays	Sigma displays
Which records belong to each group	Parent/child nodes or a collapsed group	A group outline or a collapsed group
Group counts, donut values, and location centers	The calculated group result	The same calculated group result
Which groups are open or closed	The matching visible nodes and connections	The same visible nodes and connections
Selections and saved state	The matching highlight and camera position	The matching highlight and camera position

Main tradeoffs and recommendation

Consideration	Advantage	Why
Smooth feature-rich networks	Sigma	Its custom WebGL symbols avoid repeatedly drawing node features in an added Canvas layer
Complete measured memory	Sigma	It added about 600 MiB less Chrome working memory and about 183 MiB less graphics memory
Built-in parent/child graph support	Cytoscape	The capability is already part of Cytoscape
Immediate selection and collapse	Cytoscape	Its measured response times were lower
Customization and isolation	Sigma	Its display code is separate and easier to extend
Fallback without WebGL	Cytoscape	It can continue with its Canvas display; Sigma currently recreates WebGL

Continue with Sigma as the leading option for the next MicrobeTrace network display. Keep grouping and scientific rules in MicrobeTrace so the application is not tied to either library.

Test record

- Representative performance: 20 measurements passed.
- Feature-rich performance: 10 measurements passed.
- Complete fresh-browser and graphics memory: 10 measurements passed on September 14, 2026.
- Feature, accessibility, saved-session, and WebGL-recovery checks: 5 automated tests passed.
- Memory result IDs: 2026-09-14T15-55-34-090Z through 2026-09-14T16-06-59-073Z.